Practical 4. Foundations of perception

JXH-1104 Neuroscience of Sport Performance

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Aim

To estimate the relative size of elements in four visual figures on two occasions, and to quantify the accuracy of the class's estimates.

Materials

- The figures on this page, also shown on the projector
- LINK 1 below, to the form in which estimates are recorded
- Phone, tablet, or laptop

Procedure

Each figure contains an element labelled **A** and an element labelled **B**. The size of **A** is defined as **100** (arbitrary units). For each figure, estimate the size of **B** on the same scale, as it **appears** to be: for example, 90 if B appears 10% smaller than A, or 120 if it appears 20% larger.

Estimates are made **by eye**, from a normal viewing distance. No rulers, fingers, or on-screen measurements!

Figure 1

Assume that A is 100 (a.u.), estimate the length of line **B**.

Figure 2

Assume that the diameter of the central circle **A** is 100 (a.u.), estimate the diameter of the central circle **B**.

Figure 3

Assume that bar **A** is 100 (a.u.), estimate the length of bar **B**.

Figure 4

Assume that line A is 100 (a.u.), estimate the length of line **B**.

TASK 1 · Block 1 estimates

Open the form at LINK 1, select **Block 1**, and enter one estimate for each of the four figures. Work individually, without discussion.

The class results are then displayed and discussed. Continue below only when instructed.

Continue here when instructed

TASK 2, Block 2 estimates

New information: A and B are identical in Figures 1 to 3, and in Figure 4 B is 10% longer.

Keep this in mind, but do not let it influence your estimates. In this task, still report how large each element **appears**, not how large you know it is.

Open the form again, select **Block 2**, and estimate each figure a second time.

TASK 3, Illusion scores

For each figure, calculate an **illusion score** for each block:

illusion score = estimate − true size

Reminder: the true size of B is 100 in Figures 1, 2, and 3 and 110 in Figure 4. A positive score indicates overestimation; a negative score indicates underestimation.

Points for discussion

- In which figures were the class estimates biased, and in which direction?
- How accurate were the estimates for Figure 4, in which a real difference was present?
- Did the bias change between block 1 and block 2? What does this indicate about the relation between knowledge and perception?
- Each illusion score is a within-participant difference score (Week 3). What is the advantage of expressing the data in this form?

Additional exercises

- Figure 2 is the configuration used by Witt et al. (2012) around a golf hole. Predict whether a hole surrounded by small circles or by large circles would be perceived as larger, and what effect this might have on putting.

If the form cannot not be accessed

block 1 estimates could be: Figure 1, 121; Figure 2, 108; Figure 3, 112; Figure 4, 107.